

The dilogarithm and trilogarithm on the line $\operatorname{Re} z = 1$: vertical-line closed forms, $\operatorname{Re} \leftrightarrow \operatorname{Li}_2(-y^2)$, $\operatorname{Im} \leftrightarrow \operatorname{Ti}_2(y)$

Vladimir Reshetnikov

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Store `polylog-vertical-line.wl`. Companions: `cleo-arctan-sin-chi2` (Ti_2),
`central-binomial-arcsin-ladder` (the $\operatorname{Im} \operatorname{Li}_n(1+i)$ tower).

Abstract

Reshetnikov's Math.StackExchange question 1410276 asks for $\operatorname{Re} \operatorname{Li}_2(1 \pm i\sqrt{3})$ and its imaginary companion. We place both inside a clean general picture: on the vertical line $\operatorname{Re} z = 1$ the reflection point $1 - z = -iy$ is *purely imaginary*, so the dilogarithm reflection and duplication laws collapse, giving for all $y > 0$

$$\begin{aligned}\operatorname{Re} \operatorname{Li}_2(1 + iy) &= \frac{\pi^2}{6} - \frac{1}{2} \ln(1+y^2) \ln y - \frac{\pi}{2} \arctan y - \frac{1}{4} \operatorname{Li}_2(-y^2), \\ \operatorname{Im} \operatorname{Li}_2(1 + iy) &= \frac{\pi}{4} \ln(1+y^2) - \arctan y \ln y + \operatorname{Ti}_2(y),\end{aligned}$$

where Ti_2 is the inverse-tangent integral. The *real* part rides $\operatorname{Li}_2(-y^2)$; the *imaginary* part rides $\operatorname{Ti}_2(y)$, which for $y = \tan \theta$ is pure Clausen. The MSE results are the case $y = \sqrt{3}$; $y = 1$ gives $\operatorname{Re} \operatorname{Li}_2(1 + i) = \pi^2/16$, $\operatorname{Im} \operatorname{Li}_2(1 + i) = \frac{\pi}{4} \ln 2 + G$. At weight 3 the symmetry *half-lifts*: the real part still closes at special y ($\operatorname{Re} \operatorname{Li}_3(1 + i) = \frac{35}{64} \zeta(3) + \frac{\pi^2 \ln 2}{32}$; $\operatorname{Re} \operatorname{Li}_3(1 + i\sqrt{3})$ via $\operatorname{Li}_3(\frac{1}{4})$), but the imaginary part becomes the deep Gaussian tower $\operatorname{Im} \operatorname{Li}_3(1 + i) = \mathcal{F}_3 - \frac{\pi^3}{192}$ of the central-binomial corpus, and the general real part acquires the genuine obstruction $\int_0^y \arctan^2(t)/t dt$. All identities are two-engine verified (mpmath and Wolfram) to ≥ 50 digits.

1 The problem and the key geometric fact

[MSE 1410276](#) records

$$\operatorname{Re} \operatorname{Li}_2(1 \pm i\sqrt{3}) = \frac{\pi^2}{24} - \frac{1}{2} \ln^2 2 - \frac{1}{4} \operatorname{Li}_2\left(\frac{1}{4}\right), \quad (1)$$

with $\operatorname{Im} \operatorname{Li}_2(1 + i\sqrt{3}) = \frac{\pi \ln 2}{2} + \frac{5\sqrt{3}}{72} [\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})]$ ([a/1410829](#), [a/1410705](#)). The point $1 + i\sqrt{3} = 2e^{i\pi/3}$, but the structurally decisive fact is that for any $z = 1 + iy$ the reflection point $1 - z = -iy$ is *purely imaginary*. This is exactly what makes the dilogarithm functional equations collapse.

2 The weight-2 vertical-line theorem

Theorem 1. For $y > 0$,

$$\operatorname{Re} \operatorname{Li}_2(1 + iy) = \frac{\pi^2}{6} - \frac{1}{2} \ln(1+y^2) \ln y - \frac{\pi}{2} \arctan y - \frac{1}{4} \operatorname{Li}_2(-y^2), \quad (2)$$

$$\operatorname{Im} \operatorname{Li}_2(1 + iy) = \frac{\pi}{4} \ln(1+y^2) - \arctan y \ln y + \operatorname{Ti}_2(y), \quad \operatorname{Ti}_2(y) = \operatorname{Im} \operatorname{Li}_2(iy). \quad (3)$$

Proof. The reflection law $\operatorname{Li}_2(z) + \operatorname{Li}_2(1-z) = \zeta(2) - \ln z \ln(1-z)$ at $z = 1 + iy$, $1-z = -iy$, gives $\operatorname{Li}_2(1+iy) = \zeta(2) - \ln(1+iy) \ln(-iy) - \operatorname{Li}_2(-iy)$. The duplication law $\operatorname{Li}_2(iy) + \operatorname{Li}_2(-iy) = \frac{1}{2} \operatorname{Li}_2(-y^2)$ together with $\overline{\operatorname{Li}_2(w)} = \operatorname{Li}_2(\bar{w})$ yields $\operatorname{Re} \operatorname{Li}_2(-iy) = \frac{1}{2}(\operatorname{Li}_2(iy) + \operatorname{Li}_2(-iy)) = \frac{1}{4} \operatorname{Li}_2(-y^2)$ and $\operatorname{Im} \operatorname{Li}_2(-iy) = -\operatorname{Ti}_2(y)$. Expanding $\ln(1+iy) = \frac{1}{2} \ln(1+y^2) + i \arctan y$ and $\ln(-iy) = \ln y - \frac{i\pi}{2}$ and taking real and imaginary parts gives (2)–(3). $\square$

On the line $\operatorname{Re} z = 1$ the dilogarithm thus splits into a *real* part carried by the even real dilogarithm $\operatorname{Li}_2(-y^2)$ and an *imaginary* part carried by the odd inverse-tangent integral $\operatorname{Ti}_2(y)$. Equation (1) is the case $y = \sqrt{3}$ (using $\operatorname{Li}_2(-3) = \operatorname{Li}_2(\frac{1}{4}) - \zeta(2) + 2 \ln 2 \ln \frac{2}{3}$), and the trigamma form of the imaginary part is just $\operatorname{Ti}_2(\sqrt{3}) = \frac{\pi}{6} \ln 3 + \frac{5\sqrt{3}}{72} [\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})]$ substituted into (3).

3 The Clausen form at $y = \tan \theta$

Since $1 + i \tan \theta = e^{i\theta} / \cos \theta$, Theorem 1 combines with Ramanujan's $\operatorname{Ti}_2(\tan \theta) = \theta \ln \tan \theta + \frac{1}{2} [\operatorname{Cl}_2(2\theta) + \operatorname{Cl}_2(\pi - 2\theta)]$ so that the $\theta \ln \tan \theta$ terms cancel, leaving a pure-Clausen imaginary part:

$$\operatorname{Im} \operatorname{Li}_2(1 + i \tan \theta) = -\frac{\pi}{2} \ln \cos \theta + \frac{1}{2} [\operatorname{Cl}_2(2\theta) + \operatorname{Cl}_2(\pi - 2\theta)]. \quad (4)$$

Corollary 2. $\operatorname{Re} \operatorname{Li}_2(1 + i) = \frac{\pi^2}{16}$ and $\operatorname{Im} \operatorname{Li}_2(1 + i) = \frac{\pi \ln 2}{4} + G$ (with G Catalan's constant), from $y = 1$ in Theorem 1 (using $\operatorname{Li}_2(-1) = -\pi^2/12$, $\operatorname{Ti}_2(1) = G$).

4 Weight 3: a half-lifted symmetry

At weight 3 the real/imaginary split survives only in part. The *real* part still closes at special y :

$$\operatorname{Re} \operatorname{Li}_3(1 + i) = \frac{35}{64} \zeta(3) + \frac{\pi^2 \ln 2}{32}, \quad \operatorname{Re} \operatorname{Li}_3(1 + i\sqrt{3}) = \frac{35\zeta(3) + 18 \operatorname{Li}_3(\frac{1}{4}) + 6\pi^2 \ln 2 - 24 \ln^3 2}{144}, \quad (5)$$

the second being the exact weight-3 analogue of the $\operatorname{Li}_2(\frac{1}{4})$ in (1). But the general real part does *not* close: differentiating, $\partial_y \operatorname{Re} \operatorname{Li}_3(1 + iy) = -(\operatorname{Im} \operatorname{Li}_2(1 + iy) - y \operatorname{Re} \operatorname{Li}_2(1 + iy)) / (1 + y^2)$, and integrating the Ti_2 contribution produces $\int_0^y \arctan^2(t) / t \, dt$, a genuine weight-3 inverse-tangent-square constant that is elementary only when $\arctan y$ is a rational multiple of π (as at $y = 1, \sqrt{3}$). Meanwhile the *imaginary* part is no longer elementary at all: it is the central-binomial Gaussian tower constant

$$\operatorname{Im} \operatorname{Li}_3(1 + i) = \mathcal{F}_3 - \frac{\pi^3}{192}, \quad \mathcal{F}_3 = \sqrt{2} {}_4F_3(\{\frac{1}{2}\}^4; \{\frac{3}{2}\}^3; \frac{1}{2}), \quad (6)$$

the $n = 3$ entry of the $\operatorname{Im} \operatorname{Li}_n(1 + i)$ tower (companion report **gaussian-1plusi-tower**). So on $\operatorname{Re} z = 1$ the imaginary part is precisely the deep object that the central-binomial ladder measures, while the real part remains tractable—the two halves of the polylogarithm part ways at weight 3.

5 Verification and reproducibility

Theorem 1, the Clausen form (4), Corollary 2, and the weight-3 values (5)–(6) are verified in `mpmath` and Wolfram: the general formulas at six y and the Clausen form at $\theta = \pi/k$ ($k = 3, 4, 6, 8, 12$) to $\sim 10^{-51}$; $\operatorname{Re} \operatorname{Li}_3(1+i)$ and $\operatorname{Re} \operatorname{Li}_3(1+i\sqrt{3})$ to $\sim 10^{-74}$; and $\operatorname{Im} \operatorname{Li}_3(1+i)$ matched against $\mathcal{F}_3 - \pi^3/192$ from the central-binomial store. Closed forms are in `src/PolyLog/identities/polylog-vertical-line.wl`; scripts under `src/PolyLog/tools/scratch/li2-vertical/`.